

Problem 2 :

We denote coordinates of a point G as G_x and G_y , i.e., $G = (G_x, G_y)$.

The ABC triangle is bound by the x-axis ($x=0$), the y-axis ($y=0$) and the AC diagonal with slope

$$m = \frac{C_y - A_y}{C_x - A_x} = \frac{0-1}{1-0} = -1 \text{ and the y-intercept}$$

$b=1$, so that the line equation is

$Y = -X + 1$. Since P is chosen strictly inside the triangle, the $P_x > 0$, $P_y > 0$ and

$P_x + P_y < 1$ (as $Y = -X + 1 \Rightarrow Y + X = 1$). Therefore the probability space is $(\Omega, \mathcal{F}, P)$:

$$\Omega = \{(x, y) \in \mathbb{R}^2 \mid x > 0, y > 0, x + y < 1\} \quad (\text{open interior of the triangle } ABC)$$

$$\mathcal{F} = \mathcal{B}(\Omega)$$

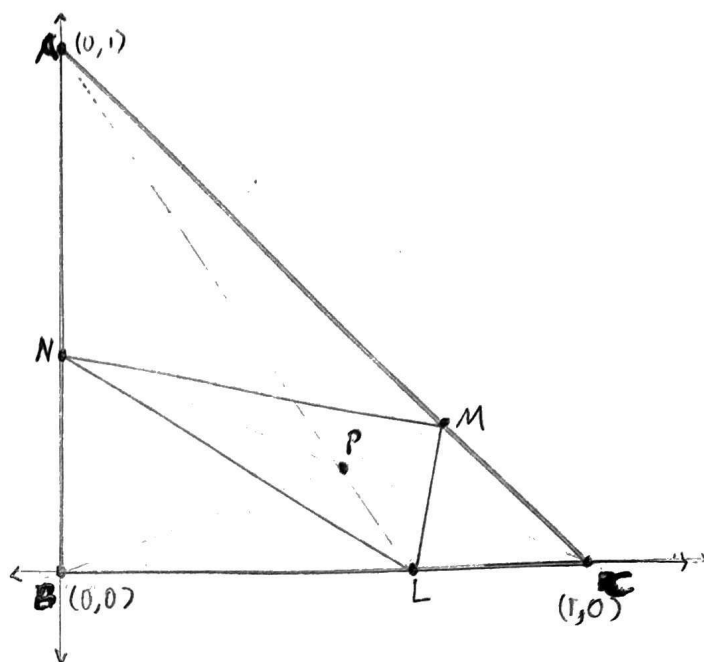
$$P(F) = \frac{\lambda(F)}{\lambda(\Omega)} \quad \forall F \in \mathcal{F}$$

Since the 2-dimensional Lebesgue measure λ coincides with the geometric area for Borel sets, which for a triangle is given by $\frac{1}{2}bh$, where b is the base and h the height w.r.t. b . For the triangle ABC, we have $b=1$ and $h=1$, and $\lambda(\Omega) = \frac{1}{2}$ and

$$P(F) = 2\lambda(F) \quad \forall F \in \mathcal{F}$$

To find the intersection points, we equate the lines passing through P (and the vertices) with the boundary lines of the triangle ABC according to the table

	Point L	Point M	Point N
Line through $P = (x, y)$	Line PA: $Y = \frac{y-1}{x} X + 1$	Line PB: $Y = \frac{y}{x} X$	Line PC: $Y = \frac{y}{x-1} X + \frac{y}{1-x}$
Intersects with	Line BC: $Y = 0$	Line AC: $-X + 1$	Line AB: $X = 0$
Intersection calculation	$0 = \frac{y-1}{x} X + 1 \Rightarrow X = \frac{x}{1-y}$	$\frac{y}{x} X = -X + 1 \Rightarrow \frac{y}{x} X + X = 1$ $\Rightarrow X(\frac{x+y}{x}) = 1 \Rightarrow X = \frac{x}{x+y}$ $Y = \frac{y}{x} (\frac{x}{x+y}) = \frac{y}{x+y}$	$Y = \frac{y}{x-1} (0) + \frac{y}{1-x}$ $\Rightarrow Y = \frac{y}{1-x}$
Final coordinate	$L = (\frac{x}{1-y}, 0)$	$M = (\frac{x}{x+y}, \frac{y}{x+y})$	$N = (0, \frac{y}{1-x})$



The lines LN , NM and LM partition the triangle ABC into four sub-triangles (BLN , ANM , CLM and LMN). These interiors of these triangles are disjoint because they only share the boundary (and the intersection of the boundary has Lebesgue measure of 0, i.e., a line has area 0). By the finite additivity of the (Lebesgue) measure, their areas must sum to the area of ΔABC . The table summarizes the calculations of the triangles BLN , ANM and CLM :

	ΔBLN	ΔANM	ΔCLM
Base Length	$L_x - B_x = \frac{x}{1-y}$	$A_y - N_y = 1 - \frac{y}{1-x} = \frac{1-x-y}{1-x}$	$C_x - L_x = 1 - \frac{x}{1-y} = \frac{1-x-y}{1-y}$
Height (wrt Base)	$N_y = \frac{y}{1-x}$	$M_x = \frac{x}{x+y}$	$M_y = \frac{y}{x+y}$
Area ($\frac{1}{2}bh$) = $\lambda(\Delta)$	$\frac{xy}{2(1-x)(1-y)}$	$\frac{x(1-x-y)}{2(1-x)(x+y)}$	$\frac{y(1-x-y)}{2(1-y)(x+y)}$

By the partition argument the area of the central triangle is

$$\begin{aligned}
 \lambda(LMN) &= \lambda(ABC) - \lambda(BLN) - \lambda(ANM) - \lambda(CLM) \\
 &= \frac{1}{2} - \frac{xy}{2(1-x)(1-y)} - \frac{x(1-x-y)}{2(1-x)(x+y)} - \frac{y(1-x-y)}{2(1-y)(x+y)} \\
 &= \frac{(1-x)(1-y)(x+y) - xy(x+y) - x(1-x-y)(1-y) - y(1-x-y)(1-x)}{2(1-x)(1-y)(x+y)} \\
 &= \frac{x+y - xy - y^2 - x^2 - xy + x^2y + xy^2 - x^2y - x^2y - x + xy + x^2 - x^2y + xy - xy^2 - y + xy + xy - x^2y + y^2 - xy^2}{2(1-x)(1-y)(x+y)} \\
 &= \frac{-2x^2y - 2xy^2 + 2xy}{2(1-x)(1-y)(x+y)} = \frac{2xy(-x-y+1)}{2(1-x)(1-y)(x+y)} = \frac{xy(1-x-y)}{(1-x)(1-y)(x+y)}
 \end{aligned}$$

We denote $E_{T>S}$ the event where the area of triangle T is strictly greater than the area of triangle S , where the areas of the triangles T and S are dependent on the randomly chosen point $P(x, y)$; i.e., $T(x, y)$ and $S(x, y)$.

$$E_{T>S} = \{(x, y) \in \Omega \mid \lambda(T(x, y)) > \lambda(S(x, y))\} = \{(x, y) \in \Omega \mid \lambda(T) > \lambda(S)\} \quad (\text{for simplicity})$$

(i) Area of $LMN >$ Area of BLN :

$$\lambda(LMN) > \lambda(BLN) \Rightarrow \frac{xy(1-x-y)}{(1-x)(1-y)(x+y)} > \frac{xy}{2(1-x)(1-y)}$$

$$(\text{since } x > 0, y > 0, 1-x > 0, 1-y > 0) \Rightarrow \frac{1-x-y}{x+y} > \frac{1}{2} \Rightarrow 2-2x-2y > x+y$$

$$\Rightarrow 2 > 3x + 3y \Rightarrow x + y < \frac{2}{3}$$

$$\text{Therefore, } E_{LMN > BLN} = \{(x, y) \in \Omega \mid x + y < \frac{2}{3}\}$$

The region forms a right triangle anchored at the origins (as $x > 0$ and $y > 0$) with legs of length $\frac{2}{3}$ along both axes, i.e.,

$$\lambda(E_{LMN} > BLN) = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{2}{3} = \frac{2}{9}$$

$$\text{and } P(E_{LMN} > BLN) = 2 \cdot \lambda(E_{LMN} > BLN) = 2 \cdot \frac{2}{9} = \frac{4}{9}$$

(ii) Area of $LMN >$ Area of AMN :

$$\lambda(LMN) > \lambda(AMN) \Leftrightarrow \frac{xy(1-x-y)}{(1-x)(1-y)(x+y)} > \frac{x(1-x-y)}{2(1-x)(x+y)}$$

$$(x > 0, y > 0, 1-x > 0, 1-y > 0) \Rightarrow \frac{y}{1-y} > \frac{1}{2} \Rightarrow 2y > 1-y \Rightarrow 3y > 1 \Rightarrow y > \frac{1}{3}$$

$$\text{Therefore, } E_{LMN} > AMN = \{(x, y) \in \Omega \mid y > \frac{1}{3}\}$$

Within the bounds of Ω ($x+y \geq 1$), the event defines a right triangle at the top of our sample space. The base is over the line $y = \frac{1}{3}$, extending from $x=0$ to $x = \frac{2}{3}$ (as $x+y = 1 \Rightarrow x = 1-y = 1-\frac{1}{3} = \frac{2}{3}$) giving a base $b = \frac{2}{3}$. The height is over $x=0$, from $y = \frac{1}{3}$ to the top vertex A, i.e., $y=1$, giving a height of $h = 1 - \frac{1}{3} = \frac{2}{3}$.

$$\lambda(E_{LMN} > AMN) = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{2}{3} = \frac{2}{9}$$

$$\text{and } P(E_{LMN} > AMN) = 2 \cdot \lambda(E_{LMN} > AMN) = 2 \cdot \frac{2}{9} = \frac{4}{9}$$